TECHNICAL PROJECT REPORT

Self-Auditing Context Engine

A Production-Ready Framework for Transparent, Explainable, and Auditable Retrieval-Augmented Generation

Semantic Retrieval

Vector search with cosine similarity and multi-factor relevance scoring across the full knowledge base

LLM-Grounded Answers

Responses strictly anchored to verified, retrieved context — never parametric memory

Full Audit Trail

Every selection and rejection logged, scored, and exportable in CSV, TXT, and JSON formats

Explainable AI

Machine scores translated into plain English reasoning accessible to non-technical stakeholders

Executive Summary & Problem Statement

THE CRISIS OF BLACK-BOX AI

> "When an AI system cannot explain its reasoning, it cannot be trusted — and it cannot be deployed responsibly."

Modern Retrieval-Augmented Generation systems have dramatically improved the factual grounding of large language models, yet they have introduced a critical governance gap: the retrieval and ranking process itself remains entirely opaque. Standard RAG pipelines silently select document chunks, weight them by opaque embedding distances, and pass them to an LLM — with no disclosure of which sources were used, why they were chosen, or what was rejected. For enterprise deployments in regulated industries, this opacity is not merely inconvenient — it is disqualifying.



⚠ Opacity

AI systems retrieve and rank documents silently, with no disclosure of which sources were used or why specific chunks were selected over others.

⚠ Unverifiability

Generated answers may blend accurate data with hallucinated content, with no mechanism to distinguish verified facts from LLM inference.

⚠ Non-Compliance

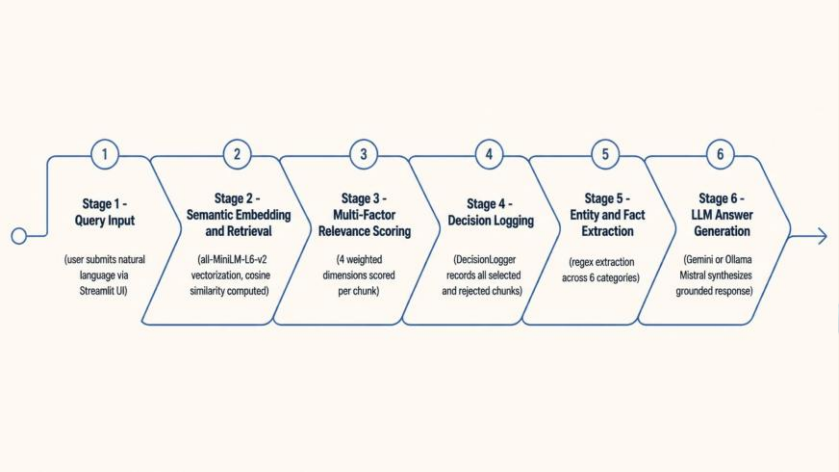
Enterprise and regulated environments require every automated decision to be logged, auditable, and challengeable by human reviewers.

The Self-Auditing Context Engine is a modular, production-ready intelligent Q&A system that solves all three problems simultaneously. Built on Retrieval-Augmented Generation (RAG), it combines semantic search, multi-factor scoring, structured entity extraction, LLM answer synthesis, and a complete machine-readable audit trail — delivering answers that are accurate, grounded, and fully accountable.

⚠ **Key Differentiator:** Unlike standard RAG systems, every retrieval decision is scored, ranked, logged, and explained — not just executed. The audit trail captures both selected and rejected chunks with full score breakdowns.

End-to-End System Architecture

The Self-Auditing Context Engine is structured as a six-stage linear pipeline, with each stage responsible for a distinct transformation of the query — from raw natural language input to a verified, explainable, and fully audited answer. The architecture is deliberately modular: each component can be independently tested, replaced, or extended without disrupting the rest of the pipeline. This design enables rapid domain adaptation, straightforward compliance reviews, and clean separation of concerns across the engineering team.



Each stage in the pipeline produces structured intermediate outputs that feed directly into the next stage, ensuring that no information is lost and every transformation is traceable. The pipeline is fully deterministic for a given knowledge base state, meaning identical queries will always produce identical retrieval results and audit trails — a critical property for reproducibility in compliance-sensitive environments.

Pipeline Stage Descriptions

01

Query Input

User submits natural language via Streamlit UI. Query is passed to the embedding layer for vectorization.

02

Semantic Embedding & Retrieval

Query encoded via all-MiniLM-L6-v2 (Sentence Transformers). Cosine similarity computed against pre-built embeddings matrix (NumPy). Similarities normalized to [0,1].

03

Multi-Factor Relevance Scoring

Each candidate chunk scored across 4 weighted dimensions: Semantic Similarity, Source Trustworthiness, Recency, and Verification Status. Configurable threshold filters results.

04

Decision Logging

DecisionLogger records every selected AND rejected chunk with full score breakdown, ranking, and human-readable reasons. Exportable as CSV.

05

Entity & Fact Extraction

EntityExtractor applies regex pattern matching across 6 categories. Facts are cross-validated against source context and tagged Verified or Derived.

06

LLM Answer Generation

Fact sheet + entity map + explainability summary passed to Google Gemini or Ollama Mistral. Response grounded strictly in retrieved, validated context.

Module Map

chunk_retriever.py
Semantic search and cosine similarity computation

decision_logger.py
Audit trail capture and CSV export

explainability_engine.py
Plain-English score translation

entity_extractor.py
Regex-based structured fact mining

llm_answer_generator.py
Gemini / Ollama Mistral integration

chunk_embedding_generator.py
Offline pre-computed 384-dim vectors

document_chunker.py
Raw document segmentation into JSON chunks

streamlit_main_app.py
Browser-based UI and pipeline orchestration

Retrieval Engine: Precision Through Multi-Factor Intelligence

The retrieval engine is the analytical core of the Self-Auditing Context Engine. Rather than relying on a single similarity metric, it applies a configurable, multi-factor scoring model that evaluates every candidate chunk across four weighted dimensions. This approach ensures that the most contextually relevant, trustworthy, and timely sources are prioritized — not merely those with the closest vector distance. The scoring model is fully transparent: every weight, threshold, and intermediate score is exposed in the audit trail.

⚡ Fast Embedding

Model: all-MiniLM-L6-v2
Dimensions: 384-dim vectors
Strategy: Offline pre-computed chunk embeddings; query-time encoding only

🎯 Cosine Similarity

Method: Normalized to [0,1]
Backend: NumPy matrix operations
Performance: Query-time encoding with pre-built similarity matrix

⚙ Configurable

Top-K: 1–10 (default: 3)
Threshold: 0.0–1.0 (default: 0.5)
Control: Per-query slider controls in UI

Multi-Factor Relevance Score Formula

$$\text{Score} = w_1 \times \text{Similarity} + w_2 \times \text{Trust} + w_3 \times \text{Recency} + w_4 \times \text{Verification}$$

Default threshold: **0.5** · All weights are fully configurable per deployment. Scores are normalized to [0,1] and exposed in the audit trail for every chunk evaluated.

Scoring Dimension Breakdown

📐 Semantic Similarity

How closely the chunk matches the query intent via vector distance in the 384-dimensional embedding space.

🛡 Trustworthiness

Source reliability score based on document metadata and origin classification within the knowledge base.

🕐 Recency

Timeliness weighting; recent documents score higher for time-sensitive queries to ensure current information is prioritized.

✅ Verification

Verified and audited documents receive a score boost; unverified sources are flagged with a warning in the audit trail.

Scoring Tier Classification

Similarity Score	Classification	Action	Rationale
> 0.90	🟢 Perfect Match	Priority Selection	Highest confidence — always included
> 0.80	🟢 Excellent	Selected	Strong semantic alignment with query
> 0.70	🟡 Strong	Selected	Relevant and well-supported
> 0.60	🟡 Good	Selected	Meets threshold with margin
> 0.50	🟠 Moderate	Conditional	Included if Top-K slots available
< 0.50	🔴 Below Threshold	Rejected + Logged	Full disqualification reason recorded

Structured Intelligence: From Raw Text to Verified Facts

The EntityExtractor and ExplainabilityEngine modules transform raw retrieved text into structured, machine-readable intelligence and human-readable reasoning. Together, they bridge the gap between opaque machine scores and the transparent, challengeable explanations that compliance officers, auditors, and business stakeholders require. The EntityExtractor operates deterministically using regex pattern matching across six predefined entity categories, ensuring consistent, reproducible extraction regardless of LLM variability.

Entity Extraction: 6 Categories

The EntityExtractor module transforms unstructured retrieved text into structured, machine-readable intelligence across six categories:

Financial Metrics

$130M, $23.4M, $145M–$155M

Percentages

18%, 12.5%, 25% YoY growth

Time Periods

Q3 2025, Q4 2025, January 2026

Dates

2025-10-15, 26/01/2026

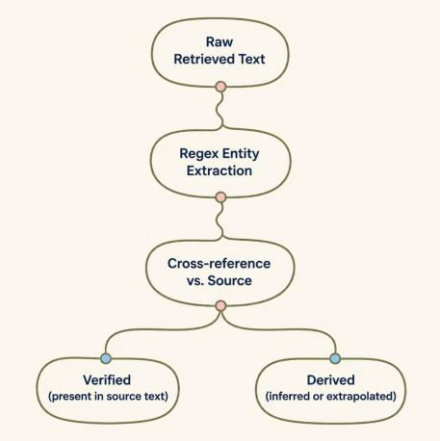
Company Names

Microsoft Corp., Google LLC

Business Metrics

Revenue, profit, margin, growth, decline

Fact Validation Workflow



Each extracted fact is cross-validated against its source context and tagged as either **Verified** (explicitly present in source text) or **Derived** (inferred or extrapolated from surrounding context). Derived facts are flagged in the audit trail for reviewer attention.

Explainability Engine Outputs

The ExplainabilityEngine translates machine scores into plain English reasoning that non-technical stakeholders can read and verify. For every query, the engine produces:

- Per-source narrative explaining why each chunk was selected or rejected
- Weighted score breakdown in readable, non-technical format
- Confidence rating with star visualization (e.g., ⭐ ⭐ ⭐ ⭐ ☆ 82.0%)
- Complete rejection log with specific disqualification reasons

Confidence Scale



Score displayed as ⭐ ⭐ ⭐ ⭐ ☆ **82.0% — Very Good**. Confidence bands are color-coded and mapped to plain-language labels for immediate stakeholder comprehension without technical interpretation.

Audit Trail: Complete Traceability at Every Step

"Every decision is logged. Every source is scored. Every rejection is explained. Nothing is hidden."

The DecisionLogger is the compliance backbone of the Self-Auditing Context Engine. It captures a complete, immutable record of every retrieval decision made during query processing — not just which chunks were selected, but precisely why every other candidate was rejected. This level of traceability is essential for enterprise governance frameworks, regulatory audits, and internal quality assurance processes. The audit trail is generated automatically for every query and is exportable in three formats to support different stakeholder workflows.

✅ Selected Chunks — Captured Data

Each selected source records:

- Chunk ID + document title + source file
- Overall confidence score (0–100%)
- Score breakdown: Similarity / Trust / Recency / Verification
- Ranked position (1st, 2nd, 3rd...)
- Up to 6 human-readable reasons for selection

Example reasons:

- "Similarity 87.3% — Strong match to query intent"
- "Verified document — Information is audited"
- "Ranked #1 — Most relevant chunk in result set"
- "Score 0.812 exceeds threshold 0.5 — Qualifies"

❌ Rejected Chunks — Captured Data

Each rejected source records:

- Chunk ID + title
- Disqualification reason with specific threshold reference
- Full composite score that failed
- Ranking position relative to Top-K window

Example reasons:

- "Score 0.38 — Below relevance threshold of 0.5"
- "Ranked #6 — Outside Top-K window (K=3)"

Rejection logging is a critical differentiator. Most RAG systems silently discard low-scoring chunks. The DecisionLogger ensures every rejection is documented and challengeable.

Audit Output Formats

CSV Audit File

Tabular log, compliance-ready. Columns: ChunkID | Score | Decision | Reason | Rank. Best for compliance teams and governance review workflows.

TXT Answer File

Natural language answer with inline source references. Best for end-user consumption and report distribution to non-technical stakeholders.

JSON Export

Complete structured result package including answer, entities, facts, audit trail, and confidence scores. Best for system integration and API consumption.

Sample Audit Trail

Chunk ID	Title	Score	Decision	Reason
C-001	Q3 Financial Report	0.912	✅ Selected	Rank #1, Score > 0.5
C-004	Market Analysis	0.834	✅ Selected	Rank #2, Verified
C-007	Strategic Outlook	0.761	✅ Selected	Rank #3, Recent
C-012	Legacy Data 2022	0.423	❌ Rejected	Below threshold
C-019	Unverified Source	0.389	❌ Rejected	Below threshold

Streamlit Interface & Knowledge Base Design

The user-facing layer of the Self-Auditing Context Engine is a fully-featured, browser-based application built on Streamlit. It delivers the complete six-stage pipeline through an intuitive, real-time interface that requires no technical expertise from the end user. The interface is designed for both interactive query sessions and batch workflows, with configurable parameters exposed through sidebar controls and a comprehensive multi-tab results panel that surfaces every layer of the system's reasoning.

UI Feature Architecture

Query Interface

- Free-text input with full natural language support
- 5 Quick Query presets: Q3 Revenue, Market Trends, Q4 Forecast, Business Growth, Strategic Goals
- Query history: last 5 queries tracked and reusable with one click

⚙ Sidebar Controls

- Top-K slider (1–10, default 3)
- Relevance threshold slider (0.0–1.0, default 0.5)
- Auto-export toggle for automated batch workflows

Real-Time Progress

- Live pipeline status: Retrieving → Scoring → Extracting → Generating
- Step-by-step progress with completion indicators
- Visual feedback for each stage of the six-stage pipeline

Results Tab Architecture

Facts

Verified and Derived facts with confidence scores

Entities

Entity tags grouped by category (6 types)

Audit

Full selection and rejection log

Sources

Sortable sources table with scores

Explanation

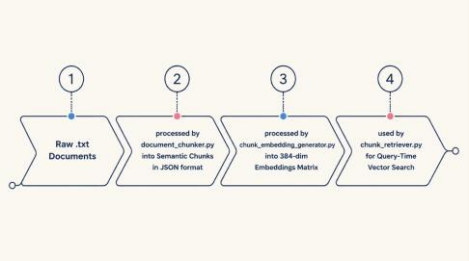
Plain-English reasoning narrative

Export

JSON / TXT / CSV download buttons

Knowledge Base Architecture

Document Ingestion Pipeline



Metadata Schema Per Chunk

```
{
  "chunk_id": "C-001",
  "source": "doc_001.txt",
  "category": "financial_report",
  "verified": true,
  "date": "Q3 2025",
  "chunk_type": "body",
  "embedding": [0.23, -0.41, ...]
// 384-dim
}
```

Knowledge base populated with financial performance documents covering quarterly revenue, regional analysis, operational KPIs, forecasts, and strategic initiatives. Each chunk carries full provenance metadata for traceability.

Outcomes, Significance & Conclusion

The Self-Auditing Context Engine represents a comprehensive solution to the governance gap that has limited responsible AI deployment in enterprise and regulated environments. By unifying semantic retrieval, multi-factor scoring, structured entity extraction, and a complete decision audit trail into a single cohesive pipeline, it delivers answers that are not merely accurate — but provably accountable. The following capability grid summarizes the six core pillars of the system.

Transparent Retrieval

Cosine similarity plus 4-factor scoring with full score visibility at every step of the pipeline. No silent filtering.

Structured Entity Extraction

Six entity types, regex-powered, with context-aware fact validation and Verified/Derived tagging.

Complete Audit Trail

CSV, TXT, and JSON exports. Selection AND rejection reasoning logged for every chunk evaluated.

Hallucination-Resistant Answers

LLM grounded strictly in verified, retrieved facts — never parametric memory or unverified inference.

Human-Readable Explainability

Every decision translated into plain English for non-technical review, challenge, and approval.

Production-Ready Interface

Streamlit UI with real-time progress, multi-format export, query history, and configurable parameters.

Technical Stack

Component	Technology	Role
Embedding Model	Sentence Transformers MiniLM-L6-v2	Query and chunk vectorization
Vector Similarity	NumPy cosine similarity	Chunk ranking and retrieval
LLM Backend	Google Gemini / Ollama Mistral	Answer generation
UI Framework	Streamlit	Web application interface
Entity Extraction	Python re (regex)	Structured fact mining
Audit Logging	JSON + CSV	Decision traceability
Export Formats	JSON, TXT, CSV	Multi-format output
Core Language	Python 3.8+	Full implementation

Strategic Significance

Enterprise Accountability

Full audit trail meets governance and compliance requirements out of the box, with no additional tooling required.

Privacy-First Option

Ollama backend enables fully local execution with zero external API dependency for sensitive data environments.

Modular Architecture

Each component is independently extensible; new domains require only knowledge base updates, not pipeline changes.

Reproducible Results

Deterministic retrieval and scoring ensure consistent, repeatable query outcomes across sessions and reviewers.

Conclusion

"The Self-Auditing Context Engine establishes a new standard for responsible AI deployment. By unifying semantic retrieval, multi-factor scoring, structured entity extraction, and a complete decision audit trail into a single cohesive pipeline, it delivers answers that are not merely accurate — but provably accountable. This architecture serves as a practical, production-ready blueprint for deploying trustworthy AI in any data-sensitive, compliance-driven environment."

Pipeline Summary: 🔍 Retrieve → 🧠 Score → 🔬 Extract → 📋 Audit → ✅ Answer